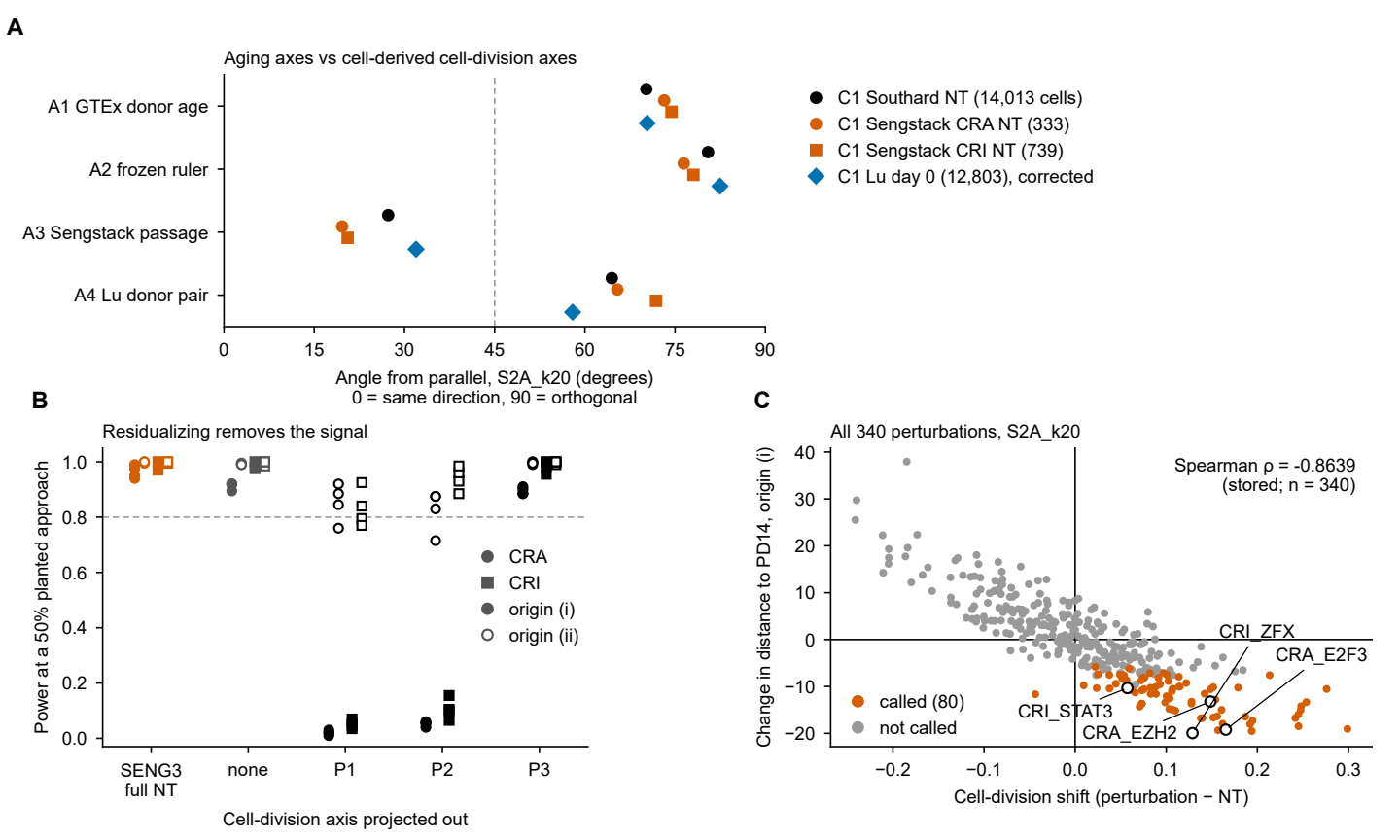

A
Aging axes vs cell-derived cell-division axes
A1 GTEx donor age
A2 frozen ruler
A3 Sengstack passage
A4 Lu donor pair
0
15
30
45
60
75
90
Angle from parallel, S2A_k20 (degrees)
0 = same direction, 90 = orthogonal
C1 Southard NT (14,013 cells)
C1 Sengstack CRA NT (333)
C1 Sengstack CRI NT (739)
C1 Lu day 0 (12,803), corrected
B
Residualizing removes the signal
Power at a 50% planted approach
1.0
0.8
0.6
0.4
0.2
0.0
CRA
CRI
origin (i)
origin (ii)
SENG3
full NT
none
P1
P2
P3
Cell-division axis projected out
C
All 340 perturbations, S2A_k20
Spearman ρ = -0.8639
(stored; n = 340)
Change in distance to PD14, origin (i)
40
30
20
10
0
−10
−20
called (80)
not called
CRI_ZFX
CRA_E2F3
CRI_STAT3
CRA_EZH2
−0.2
−0.1
0.0
0.1
0.2
0.3
Cell-division shift (perturbation − NT)